

The diagram illustrates a network architecture for connecting a Home Lab to public services. On the left, a server icon labeled "HOME LAB" is positioned behind a brick wall labeled "CGNAT". A red line labeled "VPS GATEWAY BRIDGE" connects the Home Lab to a cloud icon labeled "PUBLIC ACCESS (VPS)". On the right, a laptop and smartphone icon labeled "YOUR DEVICES" are connected to the cloud. A red wavy line labeled "PRIVATE MESH (TAILSCALE)" connects the Home Lab to the devices. Arrows labeled "PUBLIC TRAFFIC" point towards the cloud. The background features a dark blue gradient with circuit-like patterns and a star icon in the bottom right corner.

4/1/26, 12:17

misunderstanding is where most of the confusion comes from.

The Real Problem: CGNAT

Carrier-Grade NAT breaks a fundamental assumption of the internet: **you can no longer accept inbound connections.**

There's no port forwarding. No direct access. No "just open port 443".

From here, you have two fundamentally different goals:

1. **Remote access:** You → your devices.
2. **Public exposure:** Anyone → your service.

These are **not** the same thing. Choosing the wrong tool usually happens when you mix them up.

Tailscale: Excellent for Remote Access

Tailscale is built on top of WireGuard, but it adds incredible convenience:

- Automatic NAT traversal
- Identity management (SSO, ACLs)
- Seamless peer discovery
- Relay fallback (DERP)

What it does really well

- Access your home network from anywhere in the world.
- SSH into your servers without opening up ports.
- Connect distributed devices into a single, private mesh network.
- Zero-config setup (install → login → done).

Example: *Laptop* $\rightarrow$ *Tailscale* $\rightarrow$ *Home server*

No VPS required. No firewall rules. No headache.

Where Tailscale starts to fall short

Tailscale is optimized for **trusted clients**, not public services.

When you try to use it for public exposure, you run into limitations:

- **Requires client software:** You can't just share a link with your family or friends; they have to install Tailscale.
- **Limited public exposure:** Tailscale Funnel exists, but it has bandwidth limits and is primarily HTTP-focused.
- **Less routing control:** You abstract away the network layer, trading granular control for convenience.
- **Vendor dependency:** You rely on coordination servers to establish the connection network.

👉 **In short:** Tailscale is designed for *you* accessing your infrastructure, not serving the internet.

VPS Gateway: Built for Public Exposure

The VPS gateway pattern flips the model.

Instead of waiting for incoming connections, your home server proactively connects *outbound*.

- You rent a tiny, cheap VPS with a public IP.
- Your home server explicitly connects *outbound* to it via a persistent WireGuard tunnel.
- The VPS forwards incoming internet traffic into your home network.

Traffic flow:

User → VPS → WireGuard → Home server → Application

What this enables

- **Public access without opening ports:** Anyone with the URL can access your service.
- **Works with any protocol:** HTTP, TCP, SMTP — it doesn't matter.

- **Full control:** You orchestrate the routing, firewall, and TLS certificates.
- **No proprietary lock-in:** Built purely on open-source, standard tools.

The Trade-offs

- **More setup:** You have to configure the VPN, routing, and proxy.
- **Maintenance:** You now have a VPS to maintain and secure.
- **Slightly higher operational complexity.**

But in return, you get what self-hosting is all about: **Full control over your infrastructure.**

Side-by-Side Comparison

Access your own devices

- Tailscale:  Excellent
- VPS Gateway:  Overkill

Public service exposure

- Tailscale:  Limited
- VPS Gateway:  Ideal

Requires client install

- Tailscale:  Yes
- VPS Gateway:  No

Protocol support

- Tailscale:  Mostly internal/HTTP
- VPS Gateway:  Any

Setup complexity

- Tailscale:  Minimal

- VPS Gateway: Higher

Vendor independence

- **Tailscale:** ⚠️ Partial
- **VPS Gateway:** ✅ Full

Debuggability

- **Tailscale:**  Abstracted
- **VPS Gateway:**  Transparent

Real-World Scenarios

- **Accessing a NAS from your phone** → *Tailscale*
- **SSH into your homelab** → *Tailscale*
- **Internal infrastructure communication** → *Tailscale*
- **Sharing a service (like Nextcloud) with friends** → *VPS Gateway*
- **Hosting a media server publicly** → *VPS Gateway*

The Best Approach: Use Both

This is where things get interesting: **These tools are not competitors — they complement each other.**

The ultimate hybrid setup looks like this:

1. **Tailscale:** Used exclusively for admin access (SSH, RDP, Portainer dashboards) and internal tools.
2. **VPS Gateway:** Used exclusively for public-facing services (APIs, web apps, media servers).

Why this works so well

- You retain **zero-friction** access for yourself.
- You keep **full control** over public traffic.

- You avoid exposing your highly-privileged internal network and management interfaces to the internet.

This separation of concerns is clean, scalable, and secure.

What About Cloudflare Tunnel?

Cloudflare Tunnel sits somewhere in between. It offers easy setup (like Tailscale) and public exposure (like a VPS).

However, it comes with different trade-offs:

- **Proprietary technology.**
- **Traffic visibility:** Cloudflare decrypts your traffic.
- **Protocol limitations:** It strongly prefers HTTP/HTTPS.
- **Terms of Service constraints:** Routing large amounts of non-HTML data (like streaming video via Plex/Jellyfin) violates their standard ToS.

It's a valid option for basic web apps, but less ideal for a versatile, heavy-lifting homelab.

Final Thoughts

There is no “best” solution for CGNAT. There are only correct solutions for specific problems.

- If you want **simple remote access**, use Tailscale.
- If you want **public-facing services**, use a VPS gateway.
- If you want **both**, combine them.

The key insight is simple: **Don't choose tools based on popularity. Choose them based on the direction of your traffic.**

TL;DR: Tailscale = you → your network. VPS Gateway = internet → your service. They solve different problems, and the strongest homelab setups safely use both.

Homelab

Self Hosting

Networking

Tailscale

Home Server



Self-hosting enthusiast, building secure infrastructure behind CGNAT.

